

PROJECT SHOWCASE

Zajil Fiber Glass

Inventory & Order Management Platform

A production web platform built end-to-end for Zajil Fiber Glass Manufacturing — real-time inventory, order & payment tracking, barcode/QR label generation, analytics dashboards, a Progressive Web App layer, and Zebra ZP450 hardware integration.

Firebase Firestore

Firebase Auth

PWA

Barcode + QR

Zebra ZP450

Netlify



Inventory



Orders & Labels



Analytics



Hardware

PREPARED BY

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Green Spring Recycling LLC
Practical Training, Summer 2025-2026

LIVE DEPLOYMENT

zajilfiber.netlify.app

Real-time Firestore backend
Hardware-integrated print & scan
Desktop + mobile, installable PWA

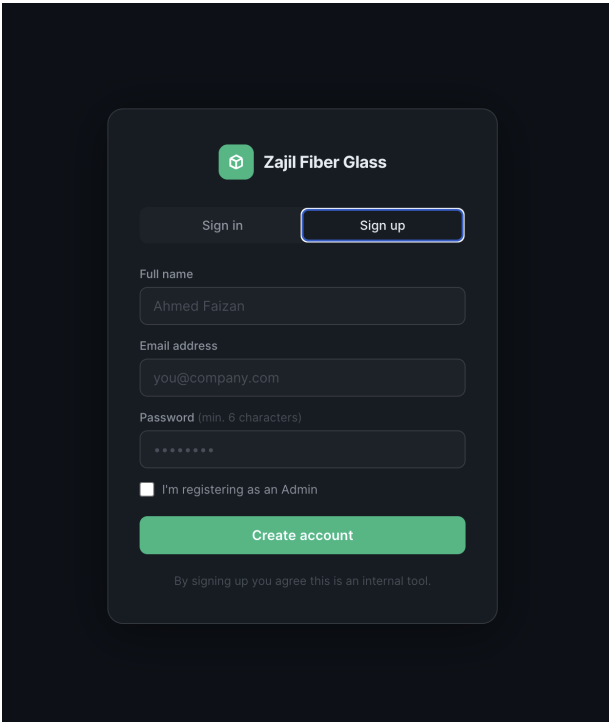
About the Project

Zajil Fiber Glass is a real-time inventory and order-management platform built independently during a Computer Engineering practical training placement at Green Spring Recycling LLC. It grew from a basic stock tracker into a complete, hardware-integrated business system covering inventory, order intake, barcode/QR label generation, revenue tracking, analytics, a Progressive Web App layer for mobile installation and offline use, an audit-history log, and a billing/invoice module.

Core Stack	Firebase Firestore - Firebase Authentication - HTML5/CSS3/JavaScript - Chart.js
Hardware	Zebra ZP450 thermal label printer - USB barcode scanner - JsBarcode - QR (qrcode.js)
Platform	Progressive Web App (manifest + Service Worker) - responsive mobile UI - Netlify CDN
Clients Served	Noon - Talabat - Deliveroo - Careem
Live URL	zajilfiber.netlify.app

Application Walkthrough

1. Authentication & Role Setup



Firebase Authentication with role-based Admin/Staff registration. Every account is tagged with a role at sign-up, driving access control across the whole platform.

2. Orders Module

Zajil Fiber Glass

Storage 17

Sold 1

Damaged 0

MAIN

Dashboard

Orders2

Add box

INVENTORY

All boxes18

In storage17

Stickers

REPORTS

Analytics

Sales & Revenue

Settings

Orders

Receive a client order → auto-generate all boxes + stickers in one click

Noon

ORD-1783108133359

03 Jul 2026, 23:48 · by faizanahmed4231@gmail.com

QTY	TYPE	REF	CONDITION	UNIT PRICE	TOTAL VALUE	PAYMENT	LOCATION
10	Normal	—	New	AED 50	AED 500	Pending	—

Print 10 stickers

View boxes

Talabat

ORD-1783104968034

03 Jul 2026, 22:56 · by faizanahmed4231@gmail.com

QTY	TYPE	REF	CONDITION	UNIT PRICE	TOTAL VALUE	PAYMENT	LOCATION
5	Large	—	New	AED 0	AED 0	Pending	Storage

Print 5 stickers

View boxes

Staff create orders for delivery-platform clients (Noon, Talabat, Deliveroo, Careem). Each order captures quantity, type, condition and unit price, and one click prints the full batch of barcode stickers for every box in that order.

3. Stickers & Barcode Labels

Stickers

Print or download stickers for any box — includes barcode, company, serial, and refurbished flag

+ Generate new barcode

QR code serial number

Export barcodes CSV

Print all

Barcode

Normal · In storage

No location set

Download PNG

View box

Barcode

Normal · In storage

No location set

Download PNG

View box

Barcode

Normal · In storage

No location set

Download PNG

View box

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View box

Barcode

Large · In storage

Storage

Download PNG

View box

Barcode

Large · In storage

Storage

Download PNG

View box

Barcode

Box · Sold / dispatched

No location set

Download PNG

View box

Barcode

Box · In storage

No location set

Download PNG

View box

Barcode

Warehouse A · Shelf 3 · down

No location set

Download PNG

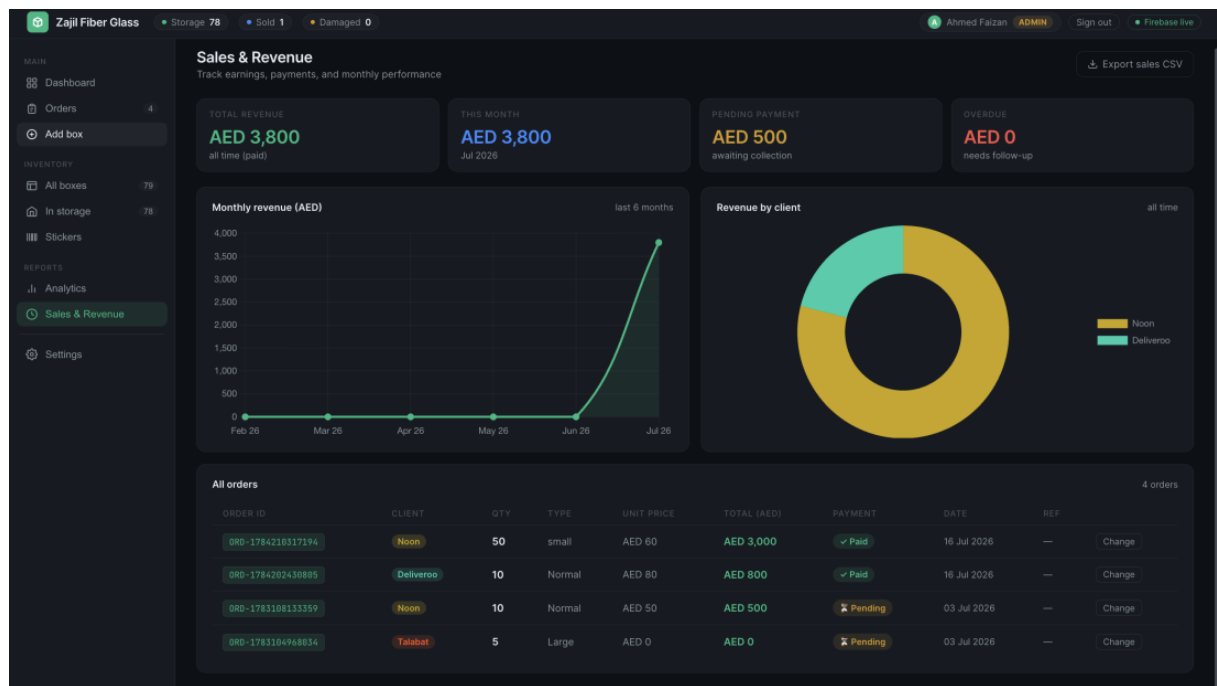
View box

Auto-generated, uniquely serialised barcode labels per box, rendered with JsBarcode. A 50-box order yields exactly 50 sequential, scannable labels ready for print or download.

Zajil Fiber Glass - Project Showcase

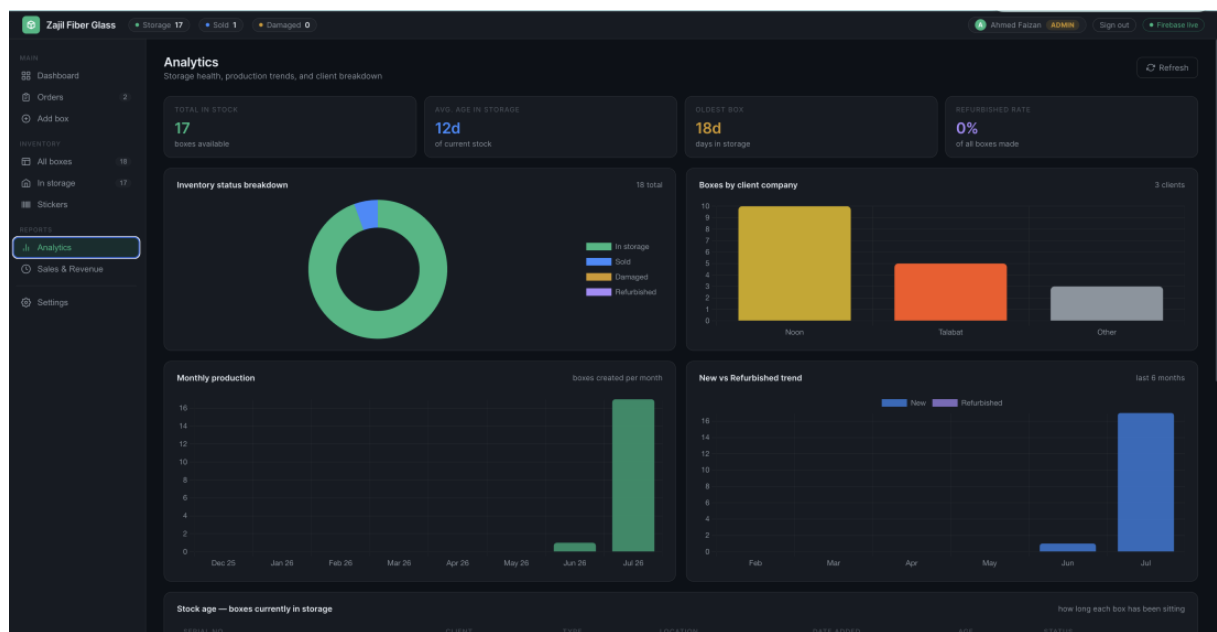
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4. Sales & Revenue Dashboard



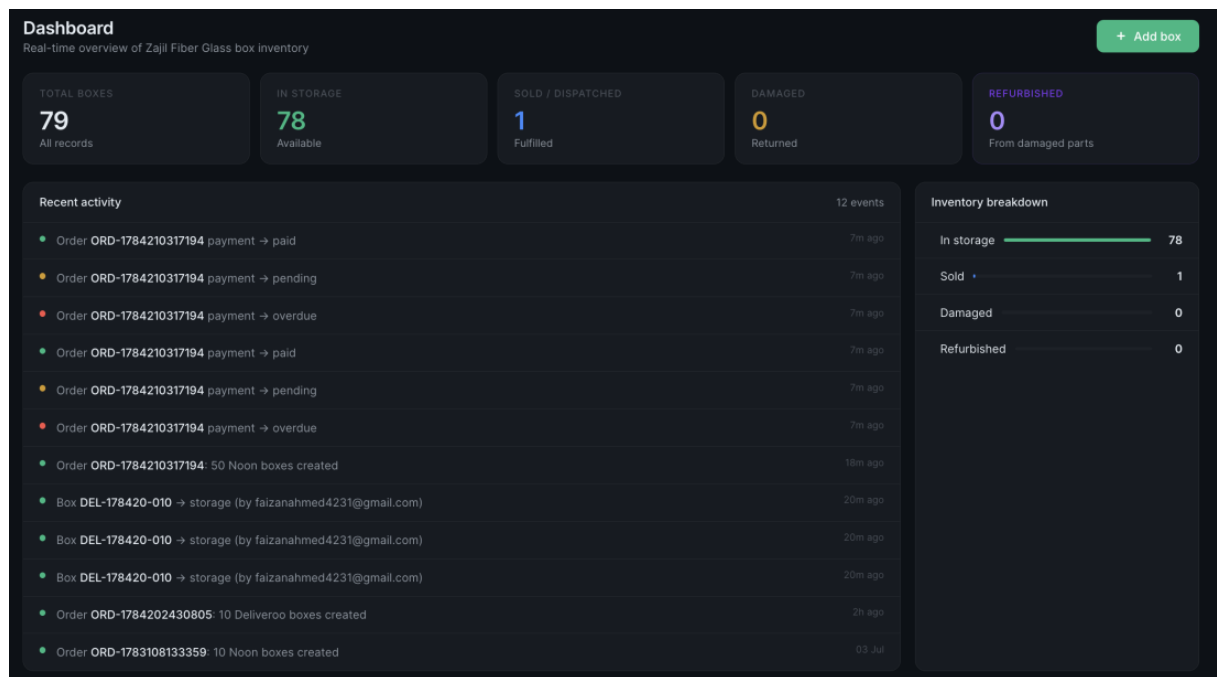
Live Firestore-driven revenue tracking: total and monthly revenue, pending payments, a per-client breakdown (Noon vs. Deliveroo vs. Talabat), and a fully exportable orders table.

5. Analytics Dashboard



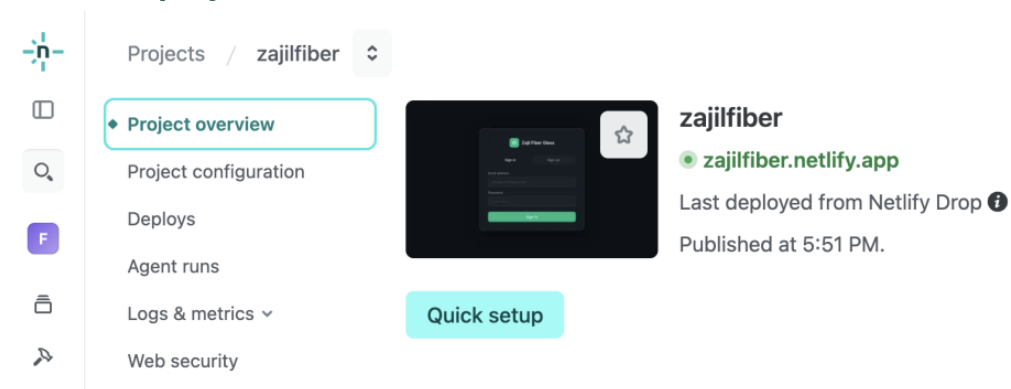
KPI cards for total stock, average box age, oldest box and refurbished rate, alongside inventory-status, per-client and monthly-production charts - all from live onSnapshot listeners.

6. Real-Time Dashboard & Activity Feed



The platform home page: live inventory summary cards plus a running activity feed that logs every order, sticker print and payment-status change with a timestamp.

7. Live Deployment



The application is deployed and continuously served from Netlify's CDN at zajilfiber.netlify.app, confirmed live and active.

8. Hardware Integration



A Zebra ZP450 thermal label printer and a USB barcode scanner are wired directly into the application, completing a fully hardware-integrated print-and-scan workflow on the warehouse floor.